

TableTime

Restaurant Reservation & Food Pre-Order Flutter Application

Cross Platform App Development (Flutter) Major Project Case Study 05

1. Introduction

Traditional restaurant booking and food ordering can require separate interactions. TableTime proposes a single flow in which a customer reserves a table and pre-orders food before arriving. This can reduce waiting at the restaurant and gives the customer a clear view of the reservation and order before checkout.

The project was implemented as a mobile-first Flutter application. The visual direction was based on the TableTime Stitch designs, while the Flutter implementation uses real widgets and application logic rather than using screenshots as UI backgrounds.

2. Problem Statement

Customers may need to spend time waiting for a table, browsing the menu after being seated, and placing an order only after arrival. A unified application can make the process more organized by allowing reservation details, table selection and food pre-ordering to be completed before the customer reaches the restaurant.

3. Objectives

- Provide a simple restaurant table reservation flow.
- Allow users to select guests, date and arrival time.
- Provide visual restaurant table selection with available, selected and occupied states.
- Display a categorized food menu using a responsive grid.
- Implement a Dart Map-based cart with quantity management.
- Calculate subtotal, service fee, tax and total dynamically.
- Provide a smooth add-to-cart animation.
- Combine reservation and food pre-order details during checkout.
- Provide a clear booking confirmation screen.
- Keep the application simple enough to understand and demonstrate during a viva.

4. Proposed System

TableTime follows a seven-screen journey: Home → Reservation Details → Table Selector → Menu → Cart → Checkout → Confirmation. Each screen has a focused purpose while the shared application state keeps reservation and cart information available throughout the journey.

Main workflow:

Home → Reservation Details → Table Selector → Menu → Cart → Checkout → Confirmation

5. Scope of the Project

The implemented prototype covers:

- Restaurant reservation details
- Restaurant table selection
- Food menu browsing and category filtering
- Food pre-ordering
- Cart quantity and removal operations
- Dynamic price calculation
- Add-to-cart animation
- Combined reservation and pre-order checkout
- Booking confirmation

Real payment processing, authentication, backend APIs, Firestore, and external calendar/wallet integrations are outside the implemented academic prototype.

6. Technology Stack

- Flutter — cross-platform mobile UI framework.
- Dart — application logic and data models.
- Material 3 — UI theme and modern Material components.
- ChangeNotifier / centralized AppState approach — simple application state management.
- Dart List and Map collections — menu/table data and cart management.
- Stitch — visual design reference and UI source of truth.

Firebase Cloud Storage was originally considered for menu image hosting but was not included in the final implementation because the required storage setup involved a paid plan. This is documented as a project limitation/future enhancement rather than an implemented feature.

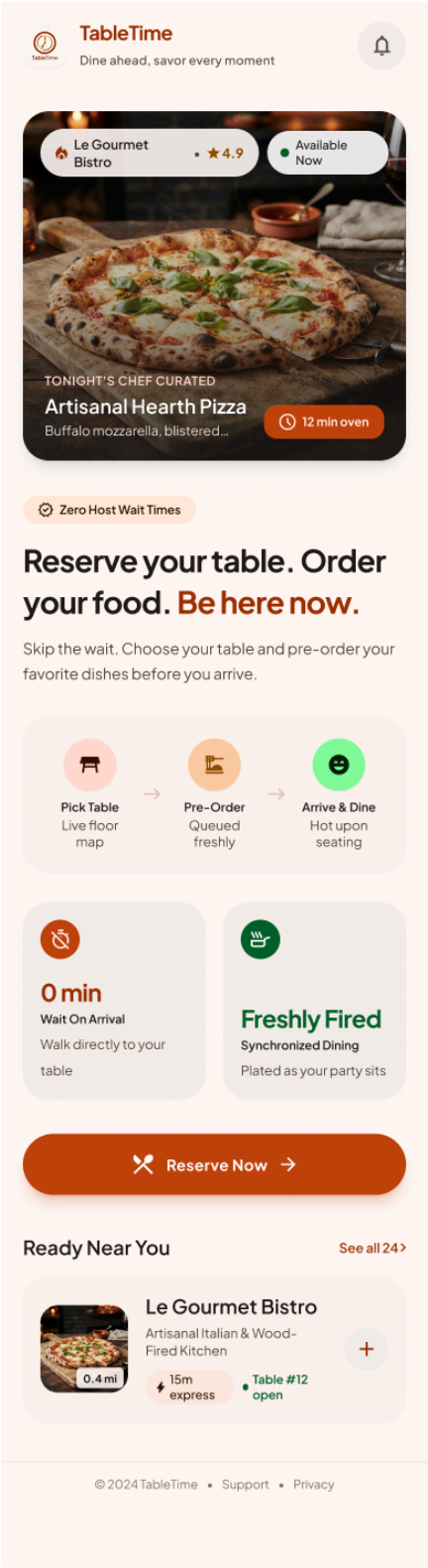
7. System Workflow

1. **Home:** Introduces TableTime and starts the reservation.
2. **Reservation Details:** Collects guest count, date and arrival time.
3. **Table Selector:** Displays restaurant tables and allows selection of available tables.
4. **Menu:** Displays dishes in categories and allows dishes to be added to the cart.
5. **Cart:** Allows quantity changes/removal and dynamically calculates pricing.
6. **Checkout:** Combines reservation information and food pre-order information.
7. **Confirmation:** Displays the final booking summary and confirmation state.

8. Application Screens

8.1 Home Screen

Shows TableTime branding, restaurant/food visual, the main reservation call-to-action and the Pick Table → Pre-Order → Arrive & Dine journey.



8.2 Reservation Details

Allows selection of party size, reservation date and arrival time and shows the reservation overview/progress.

TableTime

Restaurant Detail

1 Table

>

2 Menu

>

3 Checkout

Party Size

Standard Seating

-

2 Guests

+

Recommended for intimate dining

1

2

4

6

8+

Select Date

October 2024

Fri
09
Oct

TODAY
10
Oct

Sun
11
Oct

Mon
12
Oct

Tue
13
Oct

Arrival Time

Dinner Service

6:30 PM
Available

7:00 PM
Fast filling

7:30 PM
Prime slot

8:00 PM
Available

8:30 PM
Available

9:00 PM
Late table

YOUR PLAN OVERVIEW

✓

Instant Confirmation

Saturday, 10 Oct • 7:30 PM •
2 Guests

Main Dining Room2h allotment

●

STEP 1 OF 3

Table Selection

Choose Your Table →

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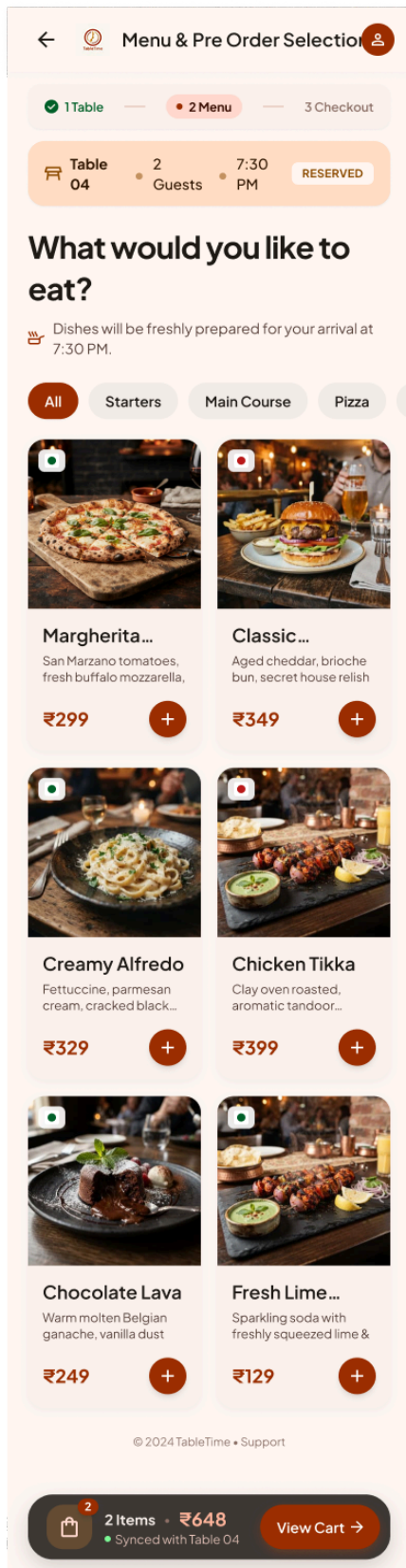
8.3 Table Selector

Displays restaurant tables with available, selected and occupied states. Only available tables can be selected.



8.4 Menu & Pre-Order

Uses category chips and a two-column food grid. Each food card shows image, name, description, price, veg/non-veg indicator and add action.



8.5 Cart

Shows selected dishes, quantities, remove controls and a dynamically calculated bill.

8.6 Checkout

Combines reservation details and food pre-order details into one review and confirmation screen.

Reservation Checkout

1 Table

2 Menu

3 Checkout

STEP 3 OF 3

Review Your Booking

Double check your table reservation and pre-ordered dishes.

Table Reservation

Table Locked & Reserved

Date

Saturday, 10 October 2024

Arrival Time

7:30 PM

Priority Seating

Guests

2 Guests

Standard seating

Selected Table

Table 04

Main Dining Room • Window View (Cap. 2)

Holding Time: 15-minute grace period upon arrival

Kitchen Pre-Order (3 items)

Edit dishes

Margherita Pizza

Wood-fired, fresh basil, mozzarella

Qty: 2

₹598.00

Classic Cheeseburger

Toasted brioche, cheddar, aioli

Qty: 1

₹349.00

Chocolate Lava Cake

Molten ganache, vanilla bean cream

Qty: 1

₹249.00

Kitchen will fire starter dishes at 7:20 PM

Payment Summary

Food Subtotal

₹1,196.00

Table Reservation & Service Fee ⓘ

₹49.00

Taxes (GST 5%)

₹59.80

Total Amount

₹1,304.80

All inclusive taxes & fees

Zero Wait Guarantee

Your table is reserved and your food will be freshly plated right when you sit.

Confirm Reservation & Pre-Order (₹1,304.80)

Cancel or Edit Selections

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8.7 Confirmation

Displays booking success, booking details, pre-order count and total amount.



TableTime



You're All Set!

Your table is reserved and your food has been thoughtfully pre-ordered.



CONFIRMED RESERVATION

Le Gourmet Bistro



TT-20458



Table
04 (Window)



Party Size
2 Guests



Date
Sat, 10 Oct



Arrival Time
7:30 PM



3 Pre-Ordered Items

Kitchen Queued

Margherita Pizza (x2), Cheeseburger (x1),
Molten Lava Cake (x1)



Authorized via UPI

₹1,304.80



Chef's Promise

We'll have everything hot and ready the moment you take your seat. Just display your reservation ticket to the reception host on arrival.



Add to Google Wallet & Calendar



Back to Home

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9. UI Design and Material 3

The application follows the Stitch visual direction with a warm cream/off-white background, burnt/deep orange primary accents, green availability/vegetarian/success states, red non-vegetarian indicators, rounded cards, rounded buttons, soft shadows and modern typography. Material 3 components are used as the Flutter foundation.

10. Data Models

1. **FoodItem:** id, name, description, price, imageUrl, isVeg, category
2. **CartItem:** foodItem, quantity
3. **RestaurantTable:** id, name, capacity, status, location
4. **Reservation:** guests, date, arrivalTime, selectedTable
5. **Booking:** reservation, cartItems, subtotal, serviceFee, tax, total

11. Cart and State Management

The cart is maintained using a Dart Map with a unique food ID as the key. When an existing dish is added, its quantity increases instead of creating a duplicate entry. Decreasing an item's quantity to zero removes it from the cart. The centralized application state stores the guest count, date, time, selected table, cart and calculated totals.

Example structure:

```
Map<String, CartItem> cart = {};
```

12. Dynamic Price Calculation

The application calculates each line total using $\text{price} \times \text{quantity}$. The subtotal is the sum of all line totals. The service fee and tax are then added to produce the final amount. Therefore, changing quantities or removing items immediately changes the displayed total rather than relying on a hardcoded bill.

Conceptual calculation:

Line Total = Item Price $\times$ Quantity

Subtotal = Sum of all Line Totals

Tax = Subtotal $\times$ Tax Rate

Total = Subtotal + Service Fee + Tax

13. Add-to-Cart Animation

When a user adds a dish, the selected food image visually moves toward the cart, shrinks/fades during the transition, and the cart responds with a short bounce/pulse. The implementation uses Flutter animation concepts such as AnimationController, Tween, CurvedAnimation, Stack, Transform and positioned/animated widgets.

14. Firebase Storage Decision

Firebase Cloud Storage was part of the original case-study requirement for menu image hosting. During final implementation, Firebase was removed because the required storage setup involved a paid plan. The final academic prototype therefore does not claim Firebase Storage integration. The menu image handling remains structured so that a future Firebase Storage service can be added without changing the main cart/reservation workflow.

15. Project Structure

```
lib/  
├── main.dart  
├── models/  
│   ├── food_item.dart  
│   ├── cart_item.dart  
│   ├── restaurant_table.dart  
│   ├── reservation.dart  
│   └── booking.dart  
├── screens/  
│   ├── home_screen.dart  
│   ├── reservation_screen.dart  
│   ├── table_selector_screen.dart  
│   ├── menu_screen.dart  
│   ├── cart_screen.dart  
│   ├── checkout_screen.dart  
│   └── confirmation_screen.dart  
├── widgets/  
├── state/  
├── data/  
└── theme/
```

16. Functional Testing

1. **Home navigation:** Tap Reserve Now → Expected: Reservation screen opens → Status: Pass
2. **Reservation:** Select guests/date/time → Expected: Selections update → Status: Pass
3. **Table selection:** Select available table → Expected: Table becomes selected → Status: Pass
4. **Occupied table:** Tap occupied table → Expected: Selection is prevented → Status: Pass
5. **Menu filter:** Select category chip → Expected: Matching dishes are displayed → Status: Pass
6. **Add to cart:** Tap add → Expected: Item, badge and totals update; animation plays → Status: Pass
7. **Duplicate item:** Add same dish twice → Expected: Quantity increases → Status: Pass
8. **Quantity:** Increase/decrease → Expected: Totals update immediately → Status: Pass
9. **Remove:** Remove cart item → Expected: Item disappears and totals update → Status: Pass
10. **Checkout:** Open checkout → Expected: Reservation + pre-order are shown together → Status: Pass
11. **Confirmation:** Confirm booking → Expected: Confirmation screen opens → Status: Pass
12. **Back to Home:** Tap Back to Home → Expected: Home screen opens and appropriate state resets → Status: Pass

17. Advantages

- Combines table reservation and food pre-ordering.
- Simple guided mobile workflow.
- Dynamic cart and pricing.
- Visual table availability.
- Category-based food browsing.
- Smooth add-to-cart interaction.
- Clear checkout and confirmation experience.

18. Limitations

- No real payment gateway.
- No user authentication.
- No restaurant backend or live restaurant API.
- Firebase Cloud Storage was not included in the final implementation because of paid-plan requirements.
- Booking data is handled within the application prototype rather than persisted in a production backend.

19. Future Scope

- Integrate Firebase Cloud Storage or another suitable image-hosting service.
- Add Firebase/other backend persistence for reservations and bookings.
- Add secure authentication.
- Add online payment integration.
- Add restaurant/admin management.
- Add booking history and notifications.
- Integrate calendar/wallet features if required in a future production version.

20. Conclusion

TableTime successfully demonstrates a complete restaurant reservation and food pre-order journey using Flutter and Dart. The application combines reservation details, restaurant table selection, menu browsing, animated cart interactions, dynamic pricing, checkout and confirmation in one flow. The project demonstrates practical use of Flutter widgets, Material 3, Dart collections, centralized state and animation. Firebase Cloud Storage was not included in the final prototype because of the paid-plan requirement, but it can be considered a future enhancement.

21. References

- Flutter and Dart documentation
- Material 3 design guidelines
- Stitch design/project used as the UI reference
- Project source code and implemented Flutter application